DOCUMENT 01

# Product Requirements Document

PRD - what we are building, for whom, and why

## Study Topic Extractor

Version: 1.0
Status: Initial Specification
Date: September 28, 2026

*Turn long study material into a focused study roadmap.*

# Table of Contents


**1. Executive Summary . . . . . 2**

1.1 Document Set and Reading Order . . . . . 2

**2. Product Vision . . . . . 3**

**3. Problem Statement . . . . . 4**

3.1 Existing Problem . . . . . 4

3.2 Proposed Solution . . . . . 4

**4. Goals and Non-Goals . . . . . 5**

4.1 Product Goals . . . . . 5

4.2 Non-Goals . . . . . 5

**5. Target Users and Personas . . . . . 6**

5.1 User Pain Points . . . . . 6

**6. User Stories and Acceptance Criteria . . . . . 7**

**7. User Journey and Core Use Cases . . . . . 8**

7.1 Primary Journey (MVP) . . . . . 8

7.2 Core Use Cases . . . . . 8

**8. Functional Requirements . . . . . 9**

8.1 Non-Functional Requirements . . . . . 9

**9. Scope: MVP, Phase 2, Future . . . . . 10**

**10. Success Metrics . . . . . 11**

10.1 Definition of MVP Success . . . . . 11

**11. Constraints, Assumptions, Limitations . . . . . 12**

11.1 Product Constraints . . . . . 12

11.2 Assumptions . . . . . 12

11.3 Limitations . . . . . 12

**12. Risks and Mitigations . . . . . 13**

# 1. Executive Summary

**Study Topic Extractor** is an AI/NLP-powered educational productivity platform that converts unstructured study material (PDFs, lecture notes, textbook chapters, transcripts, plain text) into a structured **study roadmap**: ranked topics, subtopics, keywords, short explanations, topic relationships, and exam-focused revision output.

The product is not a keyword extractor. Keywords are one input. The output is a prioritized learning map that answers one question: **"What are the most important things I should study from this document?"**

The first release (MVP) uses classical, explainable NLP (spaCy, scikit-learn, TF-IDF, n-grams) with no paid API dependency. Semantic embeddings, clustering, hierarchy, graph, Exam Focus Mode, question generation, accounts and history follow in Phase 2. LLM assistance is optional and never required for the core pipeline.

## 1.1 Document Set and Reading Order

This PRD is document 01 of 14. Terminology defined here (Topic, Keyword, Score, Priority, Analysis, Study Roadmap) is used identically in every other document. Engineering agents should read `MASTER_PROJECT_CONTEXT.pdf` first, then the document relevant to their task.

# 2. Product Vision

Every student should be able to turn hours of reading into a focused plan in under a minute. The product should feel like an intelligent study-analysis platform: fast, trustworthy, explainable, and pleasant to use on a phone the night before an exam.

- **Tagline:** Turn hours of reading into a focused study roadmap.
- **Positioning:** A general-purpose study analyzer. It is not tied to GATE, UPSC or any single exam.
- **Principle:** Explainable first. Every score can be traced to visible signals (frequency, distinctiveness, position).

# 3. Problem Statement

Students receive large volumes of material, often hundreds of pages, and must decide what matters. Manual topic extraction is slow, repetitive, inconsistent between students, and hardest for beginners who lack the subject knowledge to judge importance.

## 3.1 Existing Problem

| Current approach | Weakness |
|---|---|
| Reading everything linearly | Time-consuming; no prioritization; low retention |
| Manual highlighting / topic lists | Inconsistent; depends on prior knowledge; not reusable |
| Generic summarizers | Produce prose, not a ranked study map; no priority or relationships |
| Chatting with a general LLM | Non-deterministic, not exam-oriented, context-length limits, privacy concerns, cost |
| Keyword extraction tools | Flat word lists with no hierarchy, priority, or explanations |

## 3.2 Proposed Solution

A web application where the user pastes text or uploads a PDF and receives, within seconds, a ranked list of topics with scores, HIGH/MEDIUM/LOW study priority, keywords, extractive summaries, and export options. Phase 2 adds subtopic hierarchy, a relationship graph, Exam Focus Mode with possible questions, saved history, and progress tracking.

# 4. Goals and Non-Goals

## 4.1 Product Goals

1. G1. Produce a useful ranked topic list from a document in under 10 seconds for typical inputs.
2. G2. Make priority obvious (HIGH / MEDIUM / LOW) and understandable.
3. G3. Keep results explainable and deterministic for the same input and configuration.
4. G4. Protect private academic content (no public exposure, minimal retention).
5. G5. Be buildable by a student developer with free/open-source tools.
6. G6. Provide a clean, accessible, responsive experience in light and dark themes.

## 4.2 Non-Goals

- Payments, subscriptions, or social features (sharing, study rooms).
- Autonomous AI agents or a chat assistant in MVP or Phase 2.
- Full multilingual NLP in MVP (architecture must allow it later).
- Replacing textbooks: the product prioritizes, it does not teach.
- Guaranteeing that ranked topics equal what an examiner will ask. Output is decision support.

# 5. Target Users and Personas

| Persona | Profile | Pain points | Primary use |
|---|---|---|---|
| Aarav, engineering student | 20, semester exams, 12 PDFs of slides | Cannot tell which slide topics matter; night-before panic | Upload PDF, get HIGH topics, quick revision |
| Meera, GATE aspirant | 23, self-study, dense textbooks | Too much material, needs priority and question practice | Deep Study mode, possible questions, history |
| Rohan, placement candidate | 21, interview prep | Needs concise concept lists (DBMS, OS, CN) | Paste notes, export topic checklist |
| Dr. Iyer, teacher | Prepares revision handouts | Manual topic lists are tedious | Upload lecture notes, export PDF/CSV |

## 5.1 User Pain Points

- Information overload and unclear starting point.
- No feedback on relative importance.
- Difficulty seeing how topics relate.
- Revision time is limited; needs a compact view.
- Concern that uploaded notes are private.

# 6. User Stories and Acceptance Criteria

| ID | User story | Acceptance criteria | Phase |
|---|---|---|---|
| US-01 | As a student, I want to upload a PDF so that I can identify important topics. | PDF up to 10 MB / 200 pages accepted; text extracted; invalid or encrypted PDFs give the specified error; results shown | MVP |
| US-02 | As a student, I want to paste text so that I can analyze notes without uploading a file. | Editor accepts 200 to 150,000 chars; live character counter; Extract button disabled outside limits | MVP |
| US-03 | As a student, I want topics ranked by importance so that I know what to study first. | Topics sorted by score desc; score 0-100; rank shown; same input gives same order | MVP |
| US-04 | As a student, I want a priority level per topic so that I can plan. | HIGH >= 80, MEDIUM 50-79, LOW < 50; conveyed by label + icon + color | MVP |
| US-05 | As a student, I want topic summaries so that I can quickly understand concepts. | Each returned topic has a 1-2 sentence extractive summary drawn from the source | MVP |
| US-06 | As a student, I want keywords so that I can scan vocabulary. | Document-level keyword chips plus per-topic keywords | MVP |
| US-07 | As a student, I want to export results so that I can revise offline. | PDF, CSV, TXT downloadable (MD, JSON optional); file opens correctly | MVP |
| US-08 | As a student, I want related topics grouped so that I understand the subject structure. | Hierarchy tree with max depth 4; each topic has at most one parent | Phase 2 |
| US-09 | As a student, I want a relationship graph so that I can see connections. | Interactive graph; node size = score; color = priority; keyboard accessible | Phase 2 |
| US-10 | As a student, I want exam-focused output so that I can revise efficiently. | Quick / Normal / Deep modes show the specified content | Phase 2 |
| US-11 | As a student, I want possible questions so that I can self-test. | 3 to 6 questions per HIGH topic; each linked to a topic | Phase 2 |
| US-12 | As a returning user, I want saved documents and history so that I can revisit analyses. | Auth required; only owner can see; delete removes all data | Phase 2 |

# 7. User Journey and Core Use Cases

## 7.1 Primary Journey (MVP)

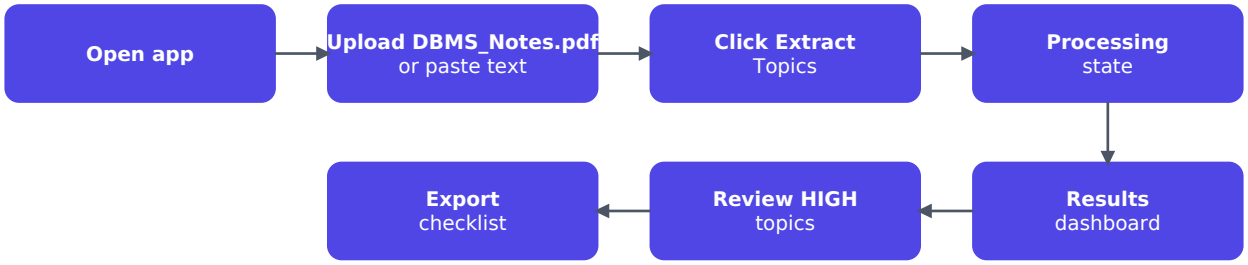



*Figure 1. MVP user journey*

Example: for `DBMS_Notes.pdf` the system identifies Functional Dependencies, Normalization, 1NF, 2NF, 3NF, BCNF, Candidate Keys, Decomposition. HIGH: Normalization, Functional Dependencies, Candidate Keys. MEDIUM: BCNF, Decomposition. LOW: supporting concepts.

## 7.2 Core Use Cases

| ID | Use case | Actor | Outcome |
|---|---|---|---|
| UC-01 | Analyze pasted notes | Any student | Ranked topics in seconds |
| UC-02 | Analyze PDF lecture slides | Any student | Ranked topics, extraction-quality warning if poor |
| UC-03 | Prepare for an exam tomorrow | Student | Quick Revision view of HIGH topics (Phase 2) |
| UC-04 | Build practice questions | Aspirant | Possible questions per topic (Phase 2) |
| UC-05 | Create handout for class | Teacher | Export PDF/CSV of topics and summaries |
| UC-06 | Return to a past analysis | Registered user | Open from history (Phase 2) |

# 8. Functional Requirements

| ID | Requirement | Phase |
|---|---|---|
| FR-01 | Accept pasted text (200 to 150,000 chars). | MVP |
| FR-02 | Accept PDF upload (10 MB, 200 pages) and extract text with PyMuPDF. | MVP |
| FR-03 | Clean and preprocess text (normalize, sentences, tokens, lemmas, stopwords). | MVP |
| FR-04 | Generate unigram, bigram, trigram candidates with multi-word preference. | MVP |
| FR-05 | Compute TF-IDF, frequency and position signals; combine into a 0-100 score with configurable weights. | MVP |
| FR-06 | Classify priority HIGH/MEDIUM/LOW using configurable thresholds. | MVP |
| FR-07 | Produce document-level keywords and per-topic keywords. | MVP |
| FR-08 | Generate extractive summaries; LLM optional. | MVP |
| FR-09 | Display analysis overview, topic cards, keywords; filter by priority; search topics. | MVP |
| FR-10 | Export PDF, CSV, TXT (and optional MD, JSON). | MVP |
| FR-11 | Sentence-Transformer embeddings, semantic relevance and clustering. | Phase 2 |
| FR-12 | Topic hierarchy and relationship graph. | Phase 2 |
| FR-13 | Exam Focus Mode (Quick, Normal, Deep) and possible-question generation. | Phase 2 |
| FR-14 | OCR for scanned PDFs. | Phase 2 |
| FR-15 | Authentication, saved documents, history, settings, study sessions. | Phase 2 |
| FR-16 | Asynchronous processing with status polling for large documents. | Phase 2 |

## 8.1 Non-Functional Requirements

| Category | Requirement |
|---|---|
| Performance | Small text analysis < 3 s; typical PDF end-to-end < 12 s (see Testing document for full table) |
| Determinism | Same input + config + pipeline version yields identical topics and scores |
| Accessibility | WCAG 2.2 AA target; keyboard operable; priority never conveyed by color alone |
| Responsiveness | Mobile, tablet, laptop, desktop, large desktop layouts |
| Privacy | MVP stores nothing server-side; uploads processed in memory; text never logged |
| Security | File validation, size limits, rate limiting, CORS allow-list, secrets in env vars |
| Reliability | Graceful, user-friendly errors; no stack traces exposed |
| Cost | Free/open-source stack; no mandatory paid API |
| Maintainability | Modular services, typed code, unit tests on scoring logic |
| Scalability | Stateless API; ready for background workers, caching, more languages |

# 9. Scope: MVP, Phase 2, Future

| MVP (achievable by one student developer) | Phase 2 | Future |
|---|---|---|
| Text input; PDF upload; PyMuPDF extraction; preprocessing; n-grams; TF-IDF; keywords; topic ranking; priority; extractive summaries; results dashboard; export (PDF, CSV, TXT); light/dark theme; stateless backend | Sentence Transformers; semantic relevance; clustering; hierarchy; relationship graph; Exam Focus Mode; question generation; OCR; Supabase auth, DB, storage; history; study sessions; settings; async jobs; LLM-hybrid summaries (optional) | Study plans; revision schedule; flashcards, quizzes, MCQs; spaced repetition; weak-topic detection; multi-document comparison; voice input; multilingual (Hindi, Gujarati); mobile app; teacher dashboard; collaborative rooms; AI study assistant |

> **SCOPE RULE.** Anything not in the MVP column must not be built until the MVP acceptance criteria pass.

# 10. Success Metrics

| Metric | Target | Measurement |
|---|---|---|
| Time to first result (typical PDF) | < 12 s p90 | Backend latency logs + frontend timing |
| Topic quality F1@10 vs gold set | MVP >= 0.55; Phase 2 >= 0.65 | scripts/evaluate.py on labeled documents |
| Top-10 accuracy (gold topics in top 10) | MVP >= 0.60; Phase 2 >= 0.70 | Evaluation script |
| Successful analysis rate | >= 97% of valid inputs | Error-code counts |
| Export success rate | >= 99% | Export endpoint status codes |
| User-rated usefulness (pilot) | >= 4.0 / 5 | In-app one-tap rating (optional) |
| Accessibility | 0 critical axe violations | Automated + manual audit |

## 10.1 Definition of MVP Success

1. A user uploads a study PDF.
2. Text is extracted automatically.
3. Important topics are identified and ranked.
4. Keywords are visible.
5. High/Medium/Low priority is shown.
6. Short topic explanations are shown.
7. Results can be exported.

Phase 2 success additionally requires topic relationships, Exam Focus Mode and saved history.

# 11. Constraints, Assumptions, Limitations

## 11.1 Product Constraints

- Free/open-source tooling; LLM APIs optional.
- Buildable by a student developer; no distributed architecture in MVP.
- English first; architecture must not block Hindi or Gujarati later.
- No payment or social features.

## 11.2 Assumptions

- Inputs are mostly English, text-based, academic material.
- Users accept that importance is inferred from document signals, not from an exam syllabus.
- Users have a modern browser; mobile Safari and Chrome are supported.
- Typical documents are 500 to 30,000 words.

## 11.3 Limitations

- Poorly written or very short notes reduce quality.
- TF-IDF alone cannot understand meaning; semantic layer arrives in Phase 2.
- Scanned PDFs need OCR (Phase 2). Diagrams and complex math notation are not analyzed.
- Ambiguous or domain-specific terminology may be missed.
- Very long unstructured documents lose precision without sectioning.

# 12. Risks and Mitigations

| Risk | Likelihood | Impact | Mitigation |
|---|---|---|---|
| Poor topic quality on some domains | Medium | High | Gold-set evaluation; tunable weights; noun-phrase and PMI filters; user feedback loop |
| Scanned or badly formatted PDFs | High | Medium | Extraction-quality score, clear warning, OCR in Phase 2 |
| Scope creep beyond MVP | High | High | Strict MVP list; agent rule: no new features without approval |
| Privacy incident with uploaded notes | Low | High | Stateless MVP, private storage, RLS, no content logging, deletion endpoints |
| Model memory too large for free hosting | Medium | Medium | MVP has no torch; Phase 2 sized deployment (2 GB) or CPU-only wheels |
| Abuse / spam uploads | Medium | Medium | Rate limits, size limits, validation, optional CAPTCHA later |
| User over-trusts scores | Medium | Medium | Explainable component breakdown; disclaimer copy |